MSMS 407 : Longitudinal Data Analysis

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Efficiency of a Cross-sectional Study relative to a Longitudinal Study under AR(1) Correlation Structure

We have to calculate

$$e = \frac{\mathrm{Var}(\hat{\beta}_L)}{\mathrm{Var}(\hat{\beta}_C)} \tag{1}$$

$$\text{where } \mathrm{Var}(\hat{\beta}_C) = \frac{\sigma^2}{\sum_{i=1}^{m}(x_{i1} - \bar{x}_1)^2} \qquad \& \qquad \mathrm{Var}(\hat{\beta}_L) = \sigma^2 \left[\left\{ \sum_{i=1}^{m} X_i' R_i^{-1} X_i \right\}^{-1} \right]_{22}.$$

Consider the repeated measurements of all the subjects to have the correlation structure of an AR(1) process *i.e.* $R_i \equiv R \; \forall i = 1(1)m$ where

$$R = ((R_{jk})) \text{ with } R_{jk} = \rho^{|j-k|}.$$

1 To obtain R^{-1}, we perform an induction on n. For $n := 1$,

$$R_1 = [1] \qquad \& \qquad \& R_1^{-1} = [1]. \tag{2}$$

For $n := 2$,

$$R_2 = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \qquad \& \qquad R_2^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}. \tag{3}$$

For $n := 3$,

$$R_3 = \begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix}.$$

We employ Gauss-Jordan elimination to obtain R_3^{-1} as follows :

$$\left[\begin{array}{ccc|ccc} 1 & \rho & \rho^2 & 1 & 0 & 0 \\ \rho & 1 & \rho & 0 & 1 & 0 \\ \rho^2 & \rho & 1 & 0 & 0 & 1 \\ \rho^3 & \rho^2 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[\text{new}R_3 = R_3 - \rho^2 R_1]{\text{new}R_2 = R_2 - \rho R_1} \left[\begin{array}{ccc|ccc} 1 & \rho & \rho^2 & 1 & 0 & 0 \\ 0 & 1-\rho^2 & \rho-\rho^3 & -\rho & 1 & 0 \\ 0 & \rho-\rho^3 & 1-\rho^4 & -\rho^2 & 0 & 1 \end{array}\right]$$

$$\text{new}R_2 = \frac{1}{1-\rho^2}R_2 \longrightarrow \left[\begin{array}{ccc|ccc} 1 & \rho & \rho^2 & 1 & 0 & 0 \\ 0 & 1 & \rho & -\frac{\rho}{1-\rho^2} & \frac{1}{1-\rho^2} & 0 \\ 0 & \rho-\rho^3 & 1-\rho^4 & -\rho^2 & 0 & 1 \end{array}\right]$$

$$\begin{array}{c}\text{new } R_1 = R_1 - \rho R_2 \\ \text{new } R_3 = R_3 - (\rho-\rho^3)R_2\end{array} \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{1-\rho^2} & -\frac{\rho}{1-\rho^2} & 0 \\ 0 & 1 & \rho & -\frac{\rho}{1-\rho^2} & \frac{1}{1-\rho^2} & 0 \\ 0 & 0 & 1-\rho^2 & 0 & -\rho & 1 \end{array}\right]$$

$$\text{new } R_3 = \frac{1}{1-\rho^2}R_3 \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{1-\rho^2} & -\frac{\rho}{1-\rho^2} & 0 \\ 0 & 1 & \rho & -\frac{\rho}{1-\rho^2} & \frac{1}{1-\rho^2} & 0 \\ 0 & 0 & 1 & 0 & -\frac{\rho}{1-\rho^2} & \frac{1}{1-\rho^2} \end{array}\right]$$

$$\text{new } R_2 = R_2 - \rho R_3 \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{1-\rho^2} & -\frac{\rho}{1-\rho^2} & 0 \\ 0 & 1 & 0 & -\frac{\rho}{1-\rho^2} & \frac{1+\rho^2}{1-\rho^2} & -\frac{\rho}{1-\rho^2} \\ 0 & 0 & 1 & 0 & -\frac{\rho}{1-\rho^2} & \frac{1}{1-\rho^2} \end{array}\right]$$

$$\therefore\ R_3^{-1} = \frac{1}{1-\rho^2}\begin{bmatrix} 1 & -\rho & 0 \\ -\rho & 1+\rho^2 & -\rho \\ 0 & -\rho & 1 \end{bmatrix}. \tag{4}$$

For $n := n$, we assume

$$\therefore\ R_n^{-1} = \frac{1}{1-\rho^2}\begin{bmatrix} 1 & -\rho & 0 & 0 & \dots & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \dots & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & -\rho & 1 \end{bmatrix}_{n\times n} = \frac{T_n}{1-\rho^2}, \text{ say.} \tag{5}$$

For $n := n + 1$,

$$R_{n+1} = \begin{bmatrix} R_n & \underset{\sim}{r} \\ \underset{\sim}{r}' & 1 \end{bmatrix}. \tag{6}$$

where

$$\underset{\sim}{r} = \begin{bmatrix} \rho^n \\ \rho^{n-1} \\ \vdots \\ \rho \end{bmatrix}.$$

Recall that,

$$\begin{bmatrix} A & B \\ B' & d \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + A^{-1}BS^{-1}B'A^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}B'A^{-1} & S^{-1} \end{bmatrix} \tag{7}$$

where

$$S = d - B'A^{-1}B.$$

Here, $A = R_n$, $B = \underset{\sim}{r}$, $d = 1$.

$$A^{-1}B = R_n^{-1}\underset{\sim}{r} = \frac{1}{1-\rho^2} \cdot T_n \underset{\sim}{r} = \frac{1}{1-\rho^2} \cdot \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \rho(1-\rho^2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \rho \end{bmatrix}.$$

$$S = 1 - \underset{\sim}{r}' R_n^{-1} \underset{\sim}{r} = 1 - \begin{bmatrix} \rho^n & \rho^{n-1} & \dots & \rho \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \rho \end{bmatrix} = 1 - \rho^2.$$

Using (7),

$$R_{n+1}^{-1} = \begin{bmatrix} R_n^{-1} + \dfrac{R_n^{-1}\underset{\sim}{r}\underset{\sim}{r}'R_n^{-1}}{S} & -\dfrac{R_n^{-1}\underset{\sim}{r}}{S} \\ -\dfrac{\underset{\sim}{r}'R_n^{-1}}{S} & \dfrac{1}{S} \end{bmatrix}$$

$$
= \begin{bmatrix} \dfrac{T_n}{S} + \dfrac{R_n^{-1}\mathfrak{r}\mathfrak{r}'R_n^{-1}}{S} & -\dfrac{R_n^{-1}\mathfrak{r}}{S} \\ -\dfrac{\mathfrak{r}'R_n^{-1}}{S} & \dfrac{1}{S} \end{bmatrix}
$$

$$
= \frac{1}{S} \begin{bmatrix} T_n + R_n^{-1}\mathfrak{r}\mathfrak{r}'R_n^{-1} & -R_n^{-1}\mathfrak{r} \\ -\mathfrak{r}'R_n^{-1} & 1 \end{bmatrix}.
$$

Now,

$$
R_n^{-1}\underset{\sim}{r}\underset{\sim}{r}'R_n^{-1} = \left(R_n^{-1}\underset{\sim}{r}\right)\cdot\left(R_n^{-1}\underset{\sim}{r}\right)' = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \rho \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & \dots & 0 & \rho \end{bmatrix} = \begin{bmatrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \rho^2 \end{bmatrix}_{n\times n}.
$$

$$
\therefore T_n + R_n^{-1}\underset{\sim}{r}\underset{\sim}{r}'R_n^{-1} = \begin{bmatrix} 1 & -\rho & 0 & 0 & \dots & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \dots & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & -\rho & 1+\rho^2 \end{bmatrix}
$$

$$
\therefore R_{n+1}^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho & 0 & 0 & \dots & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \dots & 0 & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1+\rho^2 & -\rho \\ 0 & 0 & 0 & 0 & \cdots & -\rho & 1 \end{bmatrix}
$$

Hence, by the principle of mathematical induction, we conclude that

$$
\therefore R^{-1} = \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho & 0 & 0 & \dots & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \dots & 0 & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1+\rho^2 & -\rho \\ 0 & 0 & 0 & 0 & \cdots & -\rho & 1 \end{bmatrix}_{n\times n}.
$$

2 Now we have to calculate $X_i' R^{-1} X_i$, hence $\sum_{i=1}^{m} X_i' R^{-1} X_i$. We proceed as follows.

$$X_i' R^{-1} X_i$$

$$= \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_{i1} & x_{i2} & \dots & x_{in} \end{bmatrix} \cdot \frac{1}{1-\rho^2} \begin{bmatrix} 1 & -\rho & 0 & 0 & \cdots & 0 & 0 \\ -\rho & 1+\rho^2 & -\rho & 0 & \cdots & 0 & 0 \\ 0 & -\rho & 1+\rho^2 & -\rho & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1+\rho^2 & -\rho \\ 0 & 0 & 0 & 0 & \cdots & -\rho & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & x_{i1} \\ 1 & x_{i2} \\ \vdots & \vdots \\ 1 & x_{in} \end{bmatrix}$$

$$= \frac{1}{1-\rho^2} \begin{bmatrix} 1-\rho & 1-2\rho+\rho^2 & 1-2\rho+\rho^2 & \cdots & 1-\rho \\ x_{i1}-\rho x_{i2} & (1+\rho^2)x_{i2}-\rho(x_{i1}+x_{i3}) & (1+\rho^2)x_{i3}-\rho(x_{i2}+x_{i4}) & \cdots & x_{in}-\rho x_{i,n-1} \end{bmatrix} \cdot \begin{bmatrix} 1 & x_{i1} \\ 1 & x_{i2} \\ \vdots & \vdots \\ 1 & x_{in} \end{bmatrix}$$

$$= \frac{1}{1-\rho^2} \begin{bmatrix} 2(1-\rho)+(1-2\rho+\rho^2)(n-2) & (1-\rho)(x_{i1}+x_{in})+(1-2\rho+\rho^2)\sum_{j=2}^{n-1} x_{ij} \\ (1-\rho)(x_{i1}+x_{in})+(1-2\rho+\rho^2)\sum_{j=2}^{n-1} x_{ij} & x_{i1}^2+x_{in}^2+(1+\rho)^2\sum_{j=2}^{n-1} x_{ij}^2-2\rho\sum_{j=1}^{n-1} x_{ij}x_{i,j+1} \end{bmatrix}$$

$$= \frac{1}{1-\rho^2} \begin{bmatrix} 2(1-\rho)+(1-\rho)^2(n-2) & (1-\rho)(x_{i1}+x_{in})+(1-\rho)^2\sum_{j=2}^{n-1} x_{ij} \\ (1-\rho)(x_{i1}+x_{in})+(1-\rho)^2\sum_{j=2}^{n-1} x_{ij} & x_{i1}^2+x_{in}^2+(1+\rho)^2\sum_{j=2}^{n-1} x_{ij}^2-2\rho\sum_{j=1}^{n-1} x_{ij}x_{i,j+1} \end{bmatrix}$$

$$= \begin{bmatrix} A & B_i \\ B_i & C_i \end{bmatrix}, \text{ say}$$

$$\text{where } A = \frac{2(1-\rho)+(1-\rho)^2(n-2)}{1-\rho^2} = \frac{2+(1-\rho)(n-2))}{1+\rho} = \frac{n-\rho(n-2)}{1+\rho} \ \& \tag{8}$$

$$\forall i = 1(1)m,\ B_i = \frac{1}{1-\rho^2} \cdot \left\{ (1-\rho)(x_{i1}+x_{in}) + (1-\rho)^2 \sum_{j=2}^{n-1} x_{ij} \right\},$$

$$C_i = \frac{1}{1-\rho^2} \cdot \left\{ x_{i1}^2 + x_{in}^2 + (1+\rho)^2 \sum_{j=2}^{n-1} x_{ij}^2 - 2\rho \sum_{j=1}^{n-1} x_{ij}x_{i,j+1} \right\}.$$

We can rewrite

$$x_{ij} = \overline{x}_i + x_{ij} - \overline{x}_i = \overline{x}_i + d_{ij} \tag{9}$$

where $d_{ij} = x_{ij} - \overline{x}_i$.

$$\text{Of course } \sum_{j=1}^{n} d_{ij} = 0. \tag{10}$$

Using (9),

$$\begin{aligned}
B_i &= \frac{1}{1-\rho^2} \cdot \left\{ (1-\rho)(x_{i1} + x_{in}) + (1-\rho)^2 \sum_{j=2}^{n-1} x_{ij} \right\} \\
&= \frac{1}{1-\rho^2} \cdot \left\{ (1-\rho)(\overline{x}_i + d_{i1} + \overline{x}_i + d_{in}) + (1-\rho)^2 \sum_{j=2}^{n-1} (\overline{x}_i + d_{ij}) \right\} \\
\Rightarrow (1-\rho^2)B_i &= 2(1-\rho)\overline{x}_i + (n-2)(1-\rho^2)\overline{x}_i + (1-\rho) \left\{ d_{i1} + d_{in} + (1-\rho) \sum_{j=2}^{n-1} d_{ij} \right\}
\end{aligned}$$

$$\sum_{j=1}^{n} d_{ij} = 0 \Rightarrow -(d_{i1} + d_{in}) = \sum_{j=2}^{n-1} d_{ij}. \tag{11}$$

$$\begin{aligned}
\therefore (1-\rho^2)B_i &= \overline{x}_i \left\{ 2(1-\rho) + (n-2)(1-\rho^2) \right\} + \rho(d_{i1} + d_{in}), \quad \text{using (11)} \\
\Rightarrow B_i &= \frac{1}{1+\rho} \left[\overline{x}_i \left\{ 2 + (n-2)(1-\rho) \right\} + \rho(d_{i1} + d_{in}) \right] \\
\Rightarrow \sum_{i=1}^{m} B_i &= \frac{1}{1+\rho} \left[\left\{ 2 + (n-2)(1-\rho) \right\} \sum_{i=1}^{m} \overline{x}_i + \rho \sum_{i=1}^{m} (d_{i1} + d_{in}) \right]
\end{aligned} \tag{12}$$

Again using (9),

$$\begin{aligned}
C_i &= \frac{1}{1-\rho^2} \cdot \left\{ x_{i1}^2 + x_{in}^2 + (1+\rho)^2 \sum_{j=2}^{n-1} x_{ij}^2 - 2\rho \sum_{j=1}^{n-1} x_{ij} x_{i,j+1} \right\} \\
&= \frac{1}{1-\rho^2} \cdot \left\{ (\overline{x}_i + d_{i1})^2 + (\overline{x}_i + d_{in})^2 + (1+\rho)^2 \sum_{j=2}^{n-1} (\overline{x}_i + d_{ij})^2 - 2\rho \sum_{j=1}^{n-1} (\overline{x}_i + d_{ij})(\overline{x}_i + d_{i,j+1}) \right\} \\
&= \frac{1}{1-\rho^2} \cdot \left\{ \begin{array}{l} (\overline{x}_i + d_{i1})^2 + (\overline{x}_i + d_{in})^2 + (1+\rho)^2 \sum_{j=2}^{n-1} (\overline{x}_i + d_{ij})^2 \\ \quad - 2\rho \sum_{j=1}^{n-1} \left(\overline{x}_i^2 + \overline{x}_i \cdot d_{ij} + \overline{x}_i \cdot d_{i,j+1} + d_{ij} \cdot d_{i,j+1} \right) \end{array} \right\}.
\end{aligned} \tag{13}$$

Collecting the $\overline{x}_i^2$ terms, we have

$$
\begin{aligned}
&\overline{x}_i^2 \left\{1 + 1 + (1+\rho^2)(n-2) - 2\rho(n-1)\right\} \\
&= \overline{x}_i^2 \left\{2 + n - 2 + \rho^2(n-2) - 2\rho n + 2\rho\right\} \\
&= \overline{x}_i^2 \left\{n + n\rho^2 - 2\rho^2 - 2\rho n + 2\rho\right\} \\
&= \overline{x}_i^2 \left\{n(1-\rho) - \rho n(1-\rho) + 2\rho(1-\rho)\right\} \\
&= \overline{x}_i^2 \left\{n - \rho(n-2)\right\}(1-\rho) \\
&= A(1-\rho^2)\overline{x}_i^2, \text{ using (8).}
\end{aligned} \tag{14}
$$

Collecting the d_{ij} terms, we have

$$
\begin{aligned}
&2\overline{x}_i d_{i1} + 2\overline{x}_i d_{in} + 2(1+\rho^2)\overline{x}_i \sum_{j=2}^{n-1} d_{ij} - 2\overline{x}_i\rho \sum_{j=1}^{n-1} d_{ij} - 2\overline{x}_i\rho \sum_{j=1}^{n-1} d_{i,j+1} \\
&= 2\overline{x}_i \left\{ \sum_{j=1}^{n} d_{ij} + \rho^2 \sum_{j=2}^{n-1} d_{ij} - \rho \sum_{j=1}^{n-1} d_{ij} - \rho \sum_{j=1}^{n-1} d_{i,j+1} \right\} \\
&= 2\overline{x}_i \left\{ \underbrace{\sum_{j=1}^{n} d_{ij}}_{0} + \rho^2 \underbrace{\sum_{j=2}^{n-1} d_{ij}}_{-(d_{i1}+d_{in})} - \rho \underbrace{\sum_{j=1}^{n-1} d_{ij}}_{-d_{in}} - \rho \underbrace{\sum_{j=1}^{n-1} d_{i,j+1}}_{-d_{i1}} \right\}, \text{ using (10)} \\
&= 2\overline{x}_i \left\{-\rho^2(d_{i1}+d_{in}) + \rho(d_{i1}+d_{in})\right\}
\end{aligned} \tag{15}
$$

Collecting the d_{ij} terms of higher order, we have

$$
d_{i1}^2 + d_{in}^2 + (1+\rho^2)\sum_{j=2}^{n-1} d_{ij}^2 - 2\rho \sum_{j=1}^{n-1} d_{ij} d_{i,j+1} \tag{16}
$$

$$
\sum_{j=1}^{n} d_{ij}^2 = S, \text{ say.} \tag{17}
$$

$$
\text{Then, } \sum_{j=2}^{n-1} d_{ij}^2 = S - d_{i1}^2 - d_{in}^2. \tag{18}
$$

Also,

$$
\sum_{j=1}^{n-1} (d_{ij} - d_{i,j+1})^2 = \sum_{j=1}^{n-1} d_{ij}^2 + \sum_{j=1}^{n-1} d_{i,j+1}^2 - 2\sum_{j=1}^{n-1} d_{ij} d_{i,j+1}
$$

$$\Rightarrow -2\sum_{j=1}^{n-1} d_{ij}d_{i,j+1} = \sum_{j=1}^{n-1}(d_{ij}-d_{i,j+1})^2 - \sum_{j=1}^{n-1} d_{ij}^2 - \sum_{j=1}^{n-1} d_{i,j+1}^2$$

$$= \sum_{j=1}^{n-1}(d_{ij}-d_{i,j+1})^2 - (S-d_{in}^2) - (S-d_{i1}^2), \quad \text{using (17)}$$

$$= \sum_{j=1}^{n-1}(d_{ij}-d_{i,j+1})^2 - 2S + d_{i1}^2 + d_{in}^2. \tag{19}$$

Using (18), (19) in (16), we get

$$d_{i1}^2 + d_{in}^2 + (1+\rho^2)(S-d_{i1}^2-d_{in}^2) + \rho\sum_{j=1}^{n-1}(d_{ij}-d_{i,j+1})^2 - 2\rho S + \rho d_{i1}^2 + \rho d_{in}^2$$

$$= (1+\rho^2)S - 2\rho S + \rho(1-\rho)(d_{i1}^2+d_{in}^2) + \rho\sum_{j=1}^{n-1}(d_{ij}-d_{i,j+1})^2$$

$$= (1-\rho)^2 S + \rho(1-\rho)(d_{i1}^2+d_{in}^2) + \rho\sum_{j=1}^{n-1}(d_{ij}-d_{i,j+1})^2. \tag{20}$$

Assume

$$x_{ij} = a_i + j, \quad \forall i = 1(1)m, \; j = 1(1)n \tag{21}$$

where a_i is baseline value of the covariate for $i-$th subject.

$$\therefore \overline{x}_i = \frac{1}{n}\sum_{j=1}^{n} x_{ij} = \frac{1}{n}\sum_{j=1}^{n}(a_i+j)$$

$$= \frac{1}{n}\left[na_i + \frac{n(n+1)}{2}\right]$$

$$= a_i + \frac{n+1}{2} \tag{22}$$

$$\therefore d_{ij} = x_{ij} - \overline{x}_i = j - \frac{n+1}{2}. \tag{23}$$

Consequently,

(i)

$$d_{i1} + d_{in} = \left(1 - \frac{n+1}{2}\right) + \left(n - \frac{n+1}{2}\right) = 0. \tag{24}$$

(ii)

$$d_{i1}^2 + d_{in}^2 = \left(1-\frac{n+1}{2}\right)^2 + \left(n-\frac{n+1}{2}\right)^2 = \left(\frac{1-n}{2}\right)^2 + \left(\frac{n-1}{2}\right)^2 = \frac{(n-1)^2}{2}. \tag{25}$$

(iii)

$$
\begin{aligned}
S = \sum_{j=1}^{n} d_{ij}^2 &= \sum_{j=1}^{n} \left(j - \frac{n+1}{2} \right)^2 \\
&= \sum_{j=1}^{n} j^2 - 2 \cdot \frac{n+1}{2} \sum_{j=1}^{n} j + n \left(\frac{n+1}{2} \right)^2 \\
&= \frac{n(n+1)(2n+1)}{6} - (n+1)\frac{n(n+1)}{2} + n \left(\frac{n+1}{2} \right)^2 \\
&= \frac{n(n+1)}{2} \left[\frac{2n+1}{3} - (n+1) + \frac{n+1}{2} \right] \\
&= \frac{n(n+1)}{2} \left[\frac{2n+1}{3} - \frac{n+1}{2} \right] \\
&= \frac{n(n+1)}{2} \left[\frac{4n+2-3n-3}{6} \right] \\
&= \frac{n(n+1)}{2} \left[\frac{n-1}{6} \right] \\
&= \frac{n(n^2-1)}{12}. \qquad (26)
\end{aligned}
$$

(iv)

$$
\sum_{j=1}^{n} (d_{ij} - d_{i,j+1})^2 = \sum_{j=1}^{n} \left\{ \left(j - \frac{n+1}{2} \right) - \left(j + 1 - \frac{n+1}{2} \right) \right\}^2 = \sum_{j=1}^{n-1} (-1)^2 = n - 1. \qquad (27)
$$

Using (24) in (12), we have

$$
\sum_{i=1}^{m} B_i = \frac{1}{1+\rho} \left[\{2 + (n-2)(1-\rho)\} \sum_{i=1}^{m} \overline{x}_i \right] = A \sum_{i=1}^{m} \overline{x}_i = mA\overline{x}. \qquad (28)
$$

Using (24) in (15), we have

$$
2\overline{x}_i \left\{ -\rho^2 (d_{i1} + d_{in}) + \rho (d_{i1} + d_{in}) \right\} = 0. \qquad (29)
$$

Using (25), (26), (27) in (20), we have

$$
\begin{aligned}
&(1-\rho)^2 S + \rho(1-\rho)(d_{i1}^2 + d_{in}^2) + \rho \sum_{j=1}^{n-1} (d_{ij} - d_{i,j+1})^2 \\
&= (1-\rho)^2 \cdot \frac{n(n^2-1)}{12} + \rho(1-\rho) \cdot \frac{(n-1)^2}{2} + \rho(n-1) \\
&= \frac{n-1}{12} \left\{ (1-\rho)^2 n(n+1) + 6\rho(1-\rho)(n-1) + 12\rho \right\} \\
&= \frac{n-1}{12} \left\{ (1-2\rho+\rho^2) n(n+1) + (6\rho - 6\rho^2)(n-1) + 12\rho \right\}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{n-1}{12}\left\{n(n+1) - 2\rho(n^2+n-3n+3-6) + \rho^2(n^2+n-6n+6)\right\} \\
&= \frac{n-1}{12}\left\{n(n+1) - 2\rho(n^2-2n+3) + \rho^2(n^2-5n+6)\right\} \\
&= \frac{n-1}{12}\left\{n(n+1) - 2(n-3)(n+1)\rho + (n-3)(n-2)\rho^2\right\} \\
&= \frac{n-1}{12}\cdot\gamma \qquad (30)
\end{aligned}
$$

where $\gamma = n(n+1) - 2(n-3)(n+1)\rho + (n-3)(n-2)\rho^2$. Using (14), (15) $\rightarrow$ (29), (16) $\rightarrow$ (20) $\rightarrow$ (30) in (13), we have

$$
\begin{aligned}
C_i &= \frac{1}{1-\rho^2}\left[A(1-\rho^2)\overline{x}_i^2 + \frac{n-1}{12}\cdot\gamma\right] \\
&= A\overline{x}_i^2 + \frac{n-1}{12(1-\rho^2)}\cdot\gamma \\
\Rightarrow \sum_{i=1}^{m} C_i &= A\sum_{i=1}^{m}\overline{x}_i^2 + \sum_{i=1}^{m}\frac{n-1}{12(1-\rho^2)}\gamma \\
&= A\sum_{i=1}^{m}\overline{x}_i^2 + \frac{m(n-1)}{12(1-\rho^2)}\gamma \qquad (31)
\end{aligned}
$$

Now $S_w^2 = \sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-\overline{x}_i)^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} d_{ij}^2 = \sum_{i=1}^{m} S = \frac{mn(n^2-1)}{12}$, using (26).

$$
\therefore \sum_{i=1}^{m} C_i = A\sum_{i=1}^{m}\overline{x}_i^2 + \frac{S_w^2\gamma}{n(n+1)(1-\rho^2)}.
$$

$$
\therefore \sum_{i=1}^{m} X_i' R^{-1} X_i = \begin{bmatrix} mA & \sum_{i=1}^{m} B_i \\ \sum_{i=1}^{m} B_i & \sum_{i=1}^{m} C_i \end{bmatrix}.
$$

$$
\mathrm{Var}(\hat{\beta}_L) = \text{ lower right entry of } \sigma^2\cdot\left\{\sum_{i=1}^{m} X_i' R^{-1} X_i\right\}^{-1} = \frac{mA}{(mA)\left(\sum_{i=1}^{m} C_i\right) - \left(\sum_{i=1}^{m} B_i\right)^2}.
$$

3 $S_w^2 = \sum_{i=1}^{m}\sum_{j=1}^{n}(x_{ij}-\overline{x}_i)^2$ and $S_b^2 = \sum_{i=1}^{m}(\overline{x}_i-\overline{x})^2$.

$$\text{Define } \delta = \frac{S_w^2/\{m(n-1)\}}{S_b^2/(m-1)} \quad \Rightarrow \quad S_w^2 = \frac{m(n-1)}{m-1}\,\delta\, S_b^2. \tag{32}$$

Now,

$$\begin{aligned}
\left|\sum_{i=1}^{m} X_i' R^{-1} X_i\right| &= (mA)\left(\sum_{i=1}^{m} C_i\right) - \left(\sum_{i=1}^{m} B_i\right)^2 \\
&= mA\left\{A\sum_{i=1}^{m}\overline{x}_i^2 + \frac{S_w^2\gamma}{n(n+1)(1-\rho^2)}\right\} - m^2A^2\overline{x}^2 \\
&= mA^2\left\{\sum_{i=1}^{m}\overline{x}_i^2 - m\overline{x}^2\right\} + mA\cdot\frac{S_w^2\gamma}{n(n+1)(1-\rho^2)} \\
&= mA^2S_b^2 + mA\cdot\frac{S_w^2\gamma}{n(n+1)(1-\rho^2)}.
\end{aligned}$$

$$\begin{aligned}
\therefore \operatorname{Var}(\hat{\beta}_L) &= \sigma^2\cdot\frac{mA}{mA^2S_b^2 + mA\cdot\dfrac{S_w^2\gamma}{n(n+1)(1-\rho^2)}} \\
&= \frac{\sigma^2}{AS_b^2 + \dfrac{S_w^2\gamma}{n(n+1)(1-\rho^2)}} \\
&= \frac{\sigma^2(1-\rho^2)}{A(1-\rho^2)S_b^2 + \dfrac{S_w^2\gamma}{n(n+1)}} \\
&= \frac{\sigma^2(1-\rho^2)}{A(1-\rho^2)S_b^2 + \dfrac{m(n-1)}{m-1}\,\delta\, S_b^2\cdot\dfrac{\gamma}{n(n+1)}}, \quad \text{using (32)} \\
&= \frac{\sigma^2}{S_b^2}\cdot\left\{\frac{1-\rho^2}{(1-\rho)\{n-\rho(n-2)\} + \dfrac{\gamma\delta}{n+1}}\right\}, \quad \text{using (8) and assuming } m\approx m-1 \;\&\; n\approx n-1.
\end{aligned}$$

$$\therefore \operatorname{Var}(\hat{\beta}_L) = \frac{\sigma^2}{S_b^2}\cdot\left\{\frac{1-\rho^2}{(1-\rho)\{n-\rho(n-2)\} + \dfrac{\gamma\delta}{n+1}}\right\}. \tag{33}$$

4 We have $\mathrm{Var}(\hat{\beta}_C) = \dfrac{\sigma^2}{\sum\limits_{i=1}^{m}(x_{i1} - \bar{x}_1)^2}$ where $\overline{x}_i = \dfrac{1}{m}\sum\limits_{i=1}^{m} x_{i1}$.

Remember that in cross-sectional study, we have observation from the first visit only for each subject. Under the assumption $x_{ij} = a_i + j$,

$$\overline{x}_1 = \frac{1}{m}\sum_{i=1}^{m}(a_i + 1) = \overline{a} + 1.$$

Thus,

$$x_{i1} - \overline{x}_1 = (a_i + 1) - (\overline{a} + 1) = a_i - \overline{a}.$$

$$\therefore \sum_{i=1}^{m}(x_{i1} - \bar{x}_1)^2 = \sum_{i=1}^{m}(a_i - \overline{a})^2.$$

Again, using (22),

$$\overline{x} = \frac{1}{m}\sum_{i=1}^{m}\overline{x}_i = \frac{1}{m}\sum_{i=1}^{m}\left(a_i + \frac{n+1}{2}\right)^2 = \overline{a} + \frac{n+1}{2}$$

and consequently,

$$S_b^2 = \sum_{i=1}^{m}(\overline{x}_i - \overline{x})^2 = \sum_{i=1}^{m}(a_i - \overline{a})^2.$$

$$\therefore \mathrm{Var}(\hat{\beta}_C) = \frac{\sigma^2}{S_b^2}. \tag{34}$$

5 Upon doing (33) ÷ (34), we have

$$e = \frac{1 - \rho^2}{(1-\rho)\left\{n - \rho(n-2)\right\} + \dfrac{\gamma\delta}{n+1}}.$$